# Phenotypic insufficiency of local protein representations: fibre-preserving rearrangements of human C2H2 zinc-finger arrays reorder DNA-recognition residues under an invariant 25-spectrum

Zhengyi Chen
Guangxi Key Laboratory of Drug Discovery and Optimization,
Guangxi Engineering Research Center for Pharmaceutical Molecular Screening
and Druggability Evaluation, School of Pharmacy, Guilin Medical University,
Guilin 541199, P. R. China
chenzhengyi@glmc.edu.cn

Ruqing Chen
GUT Geoservice Inc., Montreal, Québec, Canada
ruqing@hotmail.com

25 September 2026

**Abstract**

A protein representation that depends only on local sequence statistics assigns the same vector to all sequences with the same multiset of $k$-mers, the *spectral fibre*. Whether this matters depends on the phenotype. For DNA-binding C2H2 zinc-finger proteins it does. Each finger reads about three base pairs through residues at helix positions $-1$, 2, 3 and 6, and the array reads its target in order. We show that for $k$ at least the length of a finger, the *multiset* of fingers is constant on the fibre, and only their *order* can change. We also give the exact condition for a reordering: the exchanged segments must be bounded by repeated $(k-1)$-mers. The conserved TGEKP linker alone is far too short to bound one at $k = 25$. Among reviewed human proteins, 17 C2H2 zinc-finger proteins have a non-trivial 25-fibre. From 1 178 distinct fibre-mates drawn as random Eulerian trails, 883 (75%) place different recognition residues at some array position. In ZNF91, a KRAB zinc-finger protein that represses SVA and L1 retrotransposons, one fibre-mate changes the recognition residues at 17 of 31 array positions. In ZNF721 one changes 17 of 23. Every window-25 local encoder is constant on these pairs, as verified for all 1 178. Under any finger-additive recognition model, the predicted binding site differs exactly when the ordered recognition residues differ. We formalize this as a phenotypic sufficiency criterion: a representation supports a phenotype if and only if the phenotype is constant on its fibres. We then bound the error of every predictor built on a local encoder. We did not measure binding. The predicted change of specificity rests on the additive model, whose limits we discuss.

**Keywords.** C2H2 zinc finger, KRAB zinc-finger proteins, DNA-binding specificity, recognition code, spectral fibre, protein representation, sufficiency, ZNF91, ZNF721.

# 1 Introduction

Protein sequence representations are often built from local statistics: amino acid and $k$-mer composition, pseudo amino acid composition, and convolutional networks with global pooling. Such a representation is constant on the *spectral fibre* $F_k(S)$ of a sequence, the set of sequences with the same $k$-mer multiset and the same first $(k-1)$-mer [1, 2, 4]. For 854 reviewed human proteins the 10-mer spectrum does not determine the sequence. C2H2 zinc-finger proteins are 3.0% of the proteome but 40.7% of those proteins [2]. Most of them are KRAB zinc-finger proteins (KZFPs).

That a representation is blind to a set of sequences does not by itself mean that it misses anything a phenotype depends on. Most members of a fibre are not biologically meaningful [5], and a phenotype may be constant on the fibre. The question is therefore phenotype-specific. It needs a phenotype that is known to depend on sequence order and not only on local content.

DNA recognition by tandem C2H2 zinc fingers is such a phenotype. Each finger contacts roughly three base pairs. The fingers bind in antiparallel order, with the N-terminal finger at the $3'$ end of the site, and the site is read as the ordered concatenation of the fingers' sub-sites [10, 14]. KZFPs, with arrays of up to 30 or more fingers, use this to recognize specific transposable-element families [7, 8]. If a fibre-preserving rearrangement reorders the fingers, a local representation cannot distinguish proteins that the recognition code says bind different sites.

This paper makes that argument exact and applies it to the human proteome. Section 2 states the recognition model and its limits. Section 3 proves that fibres preserve the multiset of fingers and gives the condition under which they reorder it. Section 4 computes fibre-mates for all 17 human C2H2 zinc-finger proteins with a non-trivial 25-fibre, with ZNF91 and ZNF721 as case studies. Section 5 formalizes phenotypic sufficiency and bounds what any predictor on a local representation can achieve.

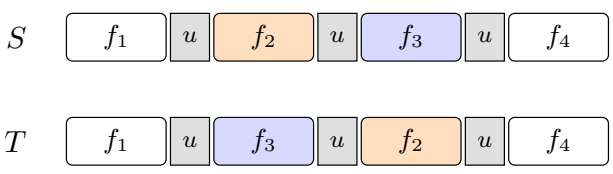
# 2 The recognition code of tandem C2H2 zinc fingers

A C2H2 zinc finger has the consensus C-$X_{2\text{–}4}$-C-$X_{12}$-H-$X_{3\text{–}5}$-H. It folds into a $\beta\beta\alpha$ unit around a zinc ion. We number the recognition helix so that the first zinc-coordinating histidine is at position +7; there is no position 0. In the Zif268 structure, residues at helix positions $-1$, 3 and 6 contact the $3'$, middle and $5'$ bases of a base triplet on the primary strand, and position 2 contacts the complementary strand [10, 6]. Consecutive fingers occupy consecutive triplets in antiparallel orientation, so the site of an array $f_1 \cdots f_n$ reads, from $5'$ to $3'$,

$$Y(S) = g(f_n)\, g(f_{n-1}) \cdots g(f_1), \tag{1}$$

where $g(f)$ is the sub-site (or the position-specific base preferences) of finger $f$. It depends on the recognition residues $r(f) = (f_{-1}, f_2, f_3, f_6)$ and, to a lesser degree, on other helix residues. We call a recognition model *finger-additive* if it has the form (1) for some map $g$. Published recognition codes predict $g$ from the helix residues, using structural rules, bacterial one-hybrid data or machine learning [11, 9].

The additive model is an approximation. Neighbouring fingers influence each other through the position-2 contact and through the linker. Not all fingers of a long KZFP array engage DNA at once, and binding also depends on chromatin [14, 7]. Our conclusions about binding are therefore conclusions about what the additive model predicts. The conclusions about representations do not depend on it.




$\mathrm{spec}_k(S) = \mathrm{spec}_k(T)$ if $|u| = k-1$ (Prop. 2); local encoder: $\phi(S) = \phi(T)$

site 5′-$g(f_4)\,g(f_3)\,g(f_2)\,g(f_1)$-3′ becomes 5′-$g(f_4)\,g(f_2)\,g(f_3)\,g(f_1)$-3′ (Eq. 1)

Figure 1: A fibre-preserving exchange of fingers. When the blocks containing $f_2$ and $f_3$ are bounded by copies of the same $(k-1)$-mer $u$ (grey), exchanging them preserves the $k$-spectrum, so every local encoder of order $k$ is unchanged. Under a finger-additive recognition model the predicted site changes whenever $g(f_2) \neq g(f_3)$.

## 3 Fibre-preserving rearrangements of finger arrays

Let $S \in \Sigma^L$ and $F_k(S) = \{T : \mathrm{spec}_k(T) = \mathrm{spec}_k(S),\ T[1..k-1] = S[1..k-1]\}$. All members of $F_k(S)$ also share the last $(k-1)$-mer of $S$ [3].

**Proposition 1** (the multiset of fingers is fibre-invariant)**.** *For $T \in F_k(S)$ and every word $w$ with $|w| \leq k$, $w$ occurs in $T$ as often as in $S$. In particular, if every finger occurrence has length at most $k$, which holds for $k \geq 24$ under the consensus above, $S$ and $T$ contain the same multiset of fingers and the same multiset of recognition-residue tuples $r(f)$.*

*Proof.* Every occurrence of $w$ starts at some position $p$. If $p \leq L-k+1$, it is the prefix of exactly one $k$-mer occurrence. Otherwise it lies in the last $(k-1)$-mer. Hence the number of occurrences of $w$ is the number of $k$-mers in $\mathrm{spec}_k(S)$ with prefix $w$, plus the number of occurrences of $w$ in the last $(k-1)$-mer. Both are the same for $T$. A finger occurrence spans 20 to 24 residues. □

So on a fibre of order $k \geq 24$ a local representation cannot lose, gain or alter a finger. It can only fail to see their arrangement. The arrangement changes by the elementary fibre moves: $S$ and $T$ in the same fibre differ by a sequence of exchanges of two adjacent segments, each beginning and ending with an occurrence of the same $(k-1)$-mer [13, 12].

**Proposition 2** (when fingers can change places)**.** *Let $S = A\,uXuY\,uB$, where $u$ is a word of length $k-1$ and $X$, $Y$ are arbitrary. Then $T = A\,uYu\,XuB$ lies in $F_k(S)$. Conversely, every $T \in F_k(S)$ arises from $S$ by a finite sequence of such exchanges. In particular, two blocks of fingers can exchange places only if they are bounded by a repeated word of length $k-1$. A linker motif of length $\ell$ that is identical between fingers bounds such an exchange only for $k \leq \ell+1$.*

*Proof.* The first part is the standard transposition argument: the $k$-mers inside $uXu$ and $uYu$ are unchanged, and the $k$-mers across each boundary read $u$ followed by the first letter of the next block, which is the same multiset in $S$ and $T$ [13]. The converse is the theorem of Ukkonen and Pevzner [13, 12]. The last statement follows because a boundary must be a repeated $(k-1)$-mer. □

The canonical KZFP linker `TGEKP` is 5 residues long and bounds exchanges only for $k \leq 6$. At $k = 25$ a boundary must be a 24-residue word that recurs, i.e. two stretches of the array that agree over most of a finger. Such near-identity arises from recent duplication of fingers within an array, the process by which KZFP arrays expand [7].

**Proposition 3** (additive models see the order)**.** *Let $T \in F_k(S)$ with $k \geq 24$, and let $r_1, \ldots, r_n$ and $r'_1, \ldots, r'_n$ be the recognition-residue tuples of the finger arrays of $S$ and $T$ in array order. For every finger-additive model whose $g$ depends only on $r$, $Y(S) = Y(T)$ whenever $r_i = r'_i$ for all $i$. If $g$ is injective on the tuples that occur, then $Y(S) \neq Y(T)$ whenever $r_i \neq r'_i$ for some $i$.*

| protein | $L$ | $\kappa$ | $\log_2 N_{25}$ | fingers | distinct mates | new order | max. positions |
|---|---|---|---|---|---|---|---|
| KLF18 | 1052 | 54 | 12.3 | 3 | 285 | 0 | 0 |
| ZNF208 | 1280 | 95 | 21.2 | 36 | 300 | 300 | 33 |
| ZNF254 | 659 | 38 | 1.0 | 13 | 1 | 1 | 2 |
| ZNF283 | 679 | 28 | 1.0 | 15 | 1 | 1 | 6 |
| ZNF347 | 839 | 35 | 2.0 | 17 | 3 | 3 | 6 |
| ZNF594 | 807 | 30 | 1.0 | 21 | 1 | 1 | 2 |
| ZNF595 | 648 | 26 | 2.0 | 17 | 3 | 3 | 9 |
| ZNF676 | 588 | 38 | 1.0 | 14 | 1 | 1 | 2 |
| ZNF709 | 641 | 34 | 4.6 | 19 | 23 | 22 | 6 |
| ZNF721 | 911 | 32 | 8.4 | 23 | 198 | 196 | 17 |
| ZNF729 | 1252 | 42 | 9.8 | 34 | 246 | 245 | 23 |
| ZNF761 | 746 | 27 | 2.0 | 17 | 3 | 3 | 11 |
| ZNF840P | 716 | 33 | 1.0 | 18 | 1 | 1 | 2 |
| ZNF888 | 718 | 26 | 1.0 | 17 | 1 | 1 | 2 |
| ZNF91 | 1191 | 51 | 5.0 | 31 | 31 | 28 | 17 |
| ZNF93 | 620 | 31 | 2.6 | 16 | 5 | 5 | 9 |
| ZNF99 | 864 | 49 | 6.4 | 21 | 75 | 72 | 9 |
| total | | | | | 1 178 | 883 | |

Table 1: Fibre-mates of order 25 of the 17 human C2H2 zinc-finger proteins with $N_{25} > 1$. "New order": mates in which some array position carries different recognition residues $(-1, 2, 3, 6)$. "Max. positions": the largest number of such array positions in one mate. In all 1 178 mates the finger multiset equals that of $S$, and the window-25 encoder output is identical to that of $S$ (relative difference below $10^{-12}$).

*Proof.* By Proposition 1 both arrays have $n$ fingers, so $Y(S)$ and $Y(T)$ are concatenations of $n$ sub-sites of equal length, compared position by position. □

Injectivity of $g$ is an idealization. Different fingers can prefer the same triplet, so the numbers below are upper bounds on the pairs for which a given code predicts different sites. They are exact counts of pairs for which the recognition residues are ordered differently.

## 4 Case studies: fibre-mates of human zinc-finger proteins

**Method.** We took all reviewed human proteins annotated with C2H2-type zinc fingers and with $N_{25} > 1$ in [1]. There are 17. For each, we drew 300 random Eulerian trails of the de Bruijn graph of order 25 starting at the protein's first 24-mer, and kept the distinct sequences $T \neq S$. These draws are not uniform on the fibre. They serve as witnesses, not as a sample. For every mate we verified $\mathrm{spec}_{25}(T) = \mathrm{spec}_{25}(S)$, parsed the fingers of $S$ and $T$ with the consensus motif, checked that the finger multisets agree (Proposition 1), compared the recognition residues position by position along the array, and applied a randomly initialized convolutional encoder of window 25 with mean pooling (float64).

**Results.** Table 1 lists the proteins. Of 1 178 distinct fibre-mates, 883 (75%) reorder the recognition residues. In every one of them the finger multiset is unchanged and the local encoder output is identical. KLF18 is an instructive exception. Its fibre is large ($\log_2 N_{25} = 12.3$), but its

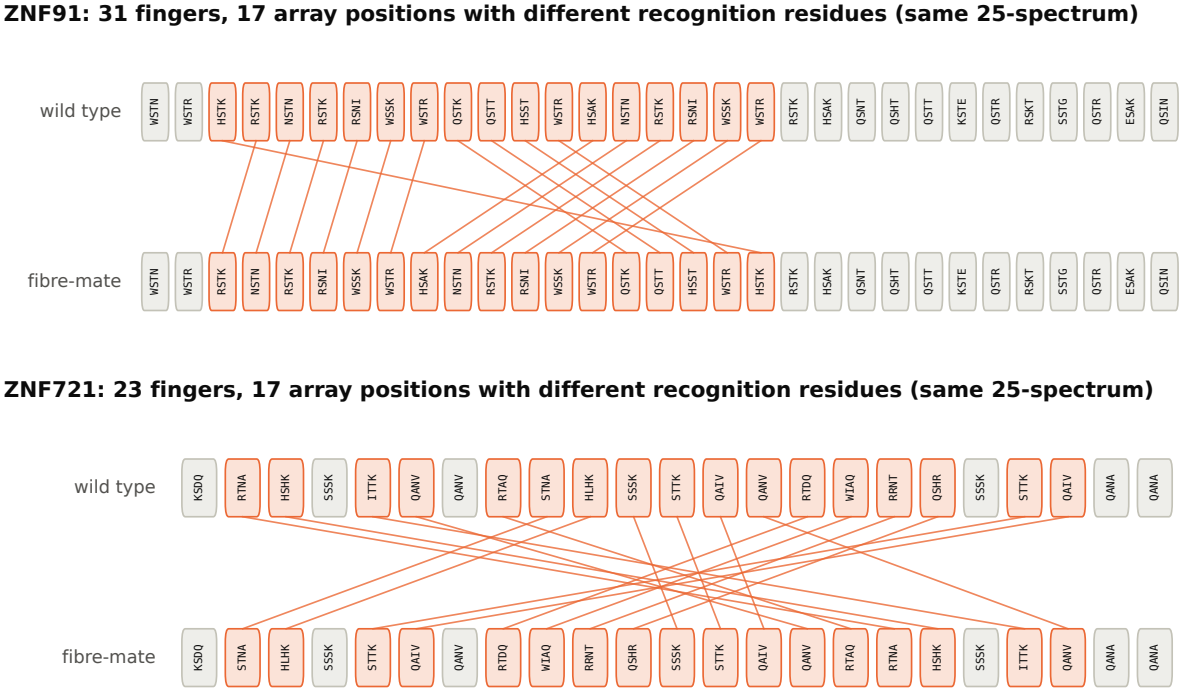



Figure 2: Fibre-preserving finger reorderings at $k = 25$. Array positions of the C2H2 fingers in the wild-type protein $S$ (top) and in a 25-fibre mate $T \in F_{25}(S)$ (bottom) for (a) ZNF91 (31 fingers; 17 positions receive a different recognition quartet) and (b) ZNF721 (23 fingers; 17 positions change). Slanted lines connect fingers that move to a new array position; unmoved fingers are unmarked. In both cases $\mathrm{spec}_{25}(S) = \mathrm{spec}_{25}(T)$ and every window-25 local encoder gives identical outputs on $S$ and $T$.

repeats lie outside its three fingers, so none of its 285 mates reorders them. For the binding-site phenotype, a local encoder of order 25 loses nothing on KLF18.

**ZNF91.** ZNF91 is a primate KZFP with 31 fingers. It binds and represses SVA and L1 retrotransposons, and its array expanded in the primate lineage [8]. Its 25-fibre has $2^{5.0}$ members. In the mate shown in Figure 2, fingers 3–19 are rearranged. Two blocks of the array exchange places across a 24-residue boundary shared by near-identical fingers, and 17 of the 31 array positions carry different recognition residues. Fingers 1–2 and 20–31 are unchanged.

Under (1), the predicted 93-bp binding site changes at 17 of its 31 triplets, while every local encoder of order 25 assigns the wild-type protein and its fibre-mate the exact same representation vector.

**ZNF721.** ZNF721 has 23 C2H2 fingers and $\log_2 N_{25} = 8.4$. Among the 198 distinct fibre-mates drawn at $k = 25$, 196 (99%) reorder the recognition-residue tuples. In the mate shown in Figure 2b, an internal block permutation across repeated 24-mer boundaries alters the recognition tuples at 17 of the 23 array positions, while preserving the finger multiset and all 25-mer counts.

## 5 Phenotypic sufficiency and predictor error bounds

We now state the general condition under which an encoder's fibres force prediction errors on a downstream phenotype. Let $\mathcal{X} \subseteq \Sigma^*$ be a domain of sequences, $\phi\colon \mathcal{X} \to \mathcal{Z}$ an encoder with fibres $F_\phi(S) = \{T \in \mathcal{X} : \phi(T) = \phi(S)\}$, and $Y\colon \mathcal{X} \to \mathcal{Y}$ a phenotype map.

**Definition 4** (phenotypic sufficiency)**.** An encoder $\phi$ is *phenotypically sufficient* for $Y$ on $\mathcal{X}$ if $Y$ is constant on every fibre of $\phi$, i.e. $\phi(S) = \phi(T)$ implies $Y(S) = Y(T)$ for all $S, T \in \mathcal{X}$.

**Theorem 5** (sufficiency criterion and irreducible error)**.** *Let $\phi\colon \mathcal{X} \to \mathcal{Z}$ and $Y\colon \mathcal{X} \to \mathcal{Y}$.*

1. *There exists a predictor $h\colon \mathcal{Z} \to \mathcal{Y}$ with $h(\phi(S)) = Y(S)$ for all $S \in \mathcal{X}$ if and only if $\phi$ is phenotypically sufficient for $Y$ on $\mathcal{X}$.*
2. *Let $d$ be a metric on $\mathcal{Y}$. For every pair $S, T \in \mathcal{X}$ in the same $\phi$-fibre and every predictor $h\colon \mathcal{Z} \to \mathcal{Y}$,*

$$\max\big(d(h(\phi(S)), Y(S)),\ d(h(\phi(T)), Y(T))\big) \ \geq\ \frac{1}{2}\, d(Y(S), Y(T)),$$

*and if $X$ is distributed uniformly on $\{S, T\}$, then $\mathbb{E}\, d(h(\phi(X)), Y(X)) \geq \frac{1}{2}\, d(Y(S), Y(T))$.*

*Proof.* (1) If $h \circ \phi = Y$, then $\phi(S) = \phi(T)$ implies $Y(S) = h(\phi(S)) = h(\phi(T)) = Y(T)$. Conversely, if $Y$ is constant on every fibre, setting $h(z) = Y(S)$ for any $S \in \phi^{-1}(z)$ is well defined and satisfies $h(\phi(S)) = Y(S)$. (2) Since $\phi(S) = \phi(T)$, both predictions equal the same point $y = h(\phi(S)) \in \mathcal{Y}$. By the triangle inequality, $d(Y(S), Y(T)) \leq d(Y(S), y) + d(y, Y(T))$, so the sum of the two errors is at least $d(Y(S), Y(T))$ and their maximum (and mean) is at least half of it. □

Combined with Proposition 3 and Section 4, Theorem 5 shows that on the 25-fibres of 16 human C2H2 zinc-finger proteins, no local encoder of order $k \leq 25$ is phenotypically sufficient for any injective finger-additive specificity model: every predictor built on $\phi$ incurs an error of at least half the motif distance between $Y(S)$ and $Y(T)$ on each reordered fibre pair.

## 6 Discussion and limitations

The gap between separating a reference set and resolving a sequence becomes a phenotypic blind spot as soon as two conditions meet: the protein's architecture places multiple domain copies between shared $(k-1)$-mer boundaries, and the biological function depends on the order in which those copies sit. Tandem C2H2 zinc-finger arrays satisfy both. At $k = 25$, the fibre preserves the exact collection of fingers (Proposition 1) while permuting blocks of fingers bounded by 24-residue repeats (Proposition 2). For 16 human zinc-finger proteins, including the retrotransposon repressors ZNF91 and ZNF93, such permutations reorder up to 17–33 recognition helices while leaving every order-25 local representation unchanged.

**Limitations.** First, we did not measure DNA binding experimentally, nor did we run structural docking. The claim that $Y(S) \neq Y(T)$ rests on the finger-additive recognition model (1). While the antiparallel ordering of triplet contacts is established by co-crystal structures and ChIP-exo footprints [10, 14, 7], only a subset of fingers in a long KZFP array may contact DNA at a given locus, so reordering non-contacting fingers would not alter binding at that locus. Second, the fibre-mates in Section 4 are mathematical Eulerian trails of $G_{25}(S)$, not documented natural

alleles; whether unequal crossing-over or ectopic gene conversion produces these specific block permutations in human populations requires long-read polymorphism data. Third, our theorems apply to local encoders. Protein language models with rotary or relative positional encodings are not local [4]; whether their learned representations separate finger-reordered mates is an empirical benchmark question.

## Data and code availability

The script `code/finger_order.py` downloads the UniProt human proteome snapshot (or reads it from cache), extracts C2H2 zinc fingers, draws fibre-mates at $k = 25$, verifies 25-spectrum and window-25 CNN invariance, and writes `results/finger_order.json`, `results/finger_order_output.txt` and `results/finger_pairs.tsv`, as well as `figures/fig_ladder.pdf`. Code, results and figures are at `https://github.com/Ruqing1963/zinc-finger-order`.

## Funding

This project was supported by the National Natural Science Foundation of China (No. 22464010), the Guangxi Natural Science Foundation of China (No. 2025GXNSFAA069294), and the 2025 Bagui Youth Top Talent Project.

## References

[1] Z. Chen, R. Chen, *What a local encoder cannot see: spectral fibres of the human proteome and the representation of perturbagens*, Zenodo (2026). doi:10.5281/zenodo.22950095

[2] Z. Chen, R. Chen, *Separation without resolution: the spectral fibre geometry of the human proteome and the blind spots of local sequence encoders*, Zenodo (2026). doi:10.5281/zenodo.22959211

[3] Z. Chen, R. Chen, *Information loss and quotient geometry of local sequence encoders: representation lattices, fibre entropy and diameter bounds*, Zenodo (2026). doi:10.5281/zenodo.22960439

[4] Z. Chen, R. Chen, *What finite-receptive-field encoders cannot distinguish: invariance theorems and architectural boundaries of convolutional sequence models*, Zenodo (2026). doi:10.5281/zenodo.22963882

[5] Z. Chen, R. Chen, *Biologically reachable spectral fibres: repeat shuffling, copy ambiguity and the scarcity of cryptic pathogenic missense variants in ClinVar*, Zenodo (2026).

[6] M. Elrod-Erickson, M. A. Rould, L. Nekludova, C. O. Pabo, *Zif268 protein–DNA complex refined at 1.6 Å: a model system for understanding zinc finger–DNA interactions*, Structure **4** (1996) 1171–1180. doi:10.1016/S0969-2126(96)00125-6

[7] M. Imbeault, P.-Y. Helleboid, D. Trono, *KRAB zinc-finger proteins contribute to the evolution of gene regulatory networks*, Nature **543** (2017) 550–554. doi:10.1038/nature21683

[8] F. M. J. Jacobs et al., *An evolutionary arms race between KRAB zinc-finger genes ZNF91/93 and SVA/L1 retrotransposons*, Nature **516** (2014) 242–245. doi:10.1038/nature13760

[9] H. S. Najafabadi et al., *C2H2 zinc finger proteins greatly expand the human regulatory lexicon*, Nat. Biotechnol. **33** (2015) 555–562. doi:10.1038/nbt.3128

[10] N. P. Pavletich, C. O. Pabo, *Zinc finger–DNA recognition: crystal structure of a Zif268–DNA complex at 2.1 Å*, Science **252** (1991) 809–817. doi:10.1126/science.2028256

[11] A. V. Persikov, M. Singh, *De novo prediction of DNA-binding specificities for C2H2 zinc finger proteins*, Nucleic Acids Res. **42** (2014) 97–108. doi:10.1093/nar/gkt890

[12] P. A. Pevzner, *DNA physical mapping and alternating Eulerian cycles in colored graphs*, Algorithmica **13** (1995) 77–105. doi:10.1007/bf01188582

[13] E. Ukkonen, *Approximate string-matching with q-grams and maximal matches*, Theoret. Comput. Sci. **92** (1992) 191–211. doi:10.1016/0304-3975(92)90143-4

[14] S. A. Wolfe, L. Nekludova, C. O. Pabo, *DNA recognition by* $Cys_2His_2$ *zinc finger proteins*, Annu. Rev. Biophys. Biomol. Struct. **29** (2000) 183–212. doi:10.1146/annurev.biophys.29.1.183